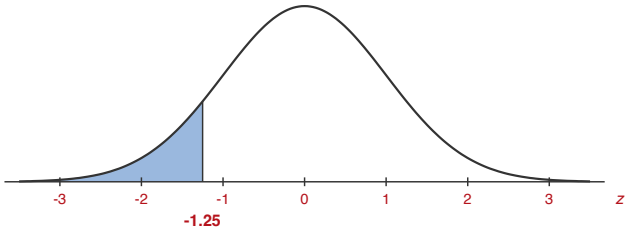
Standard Normal Areas Practice

Find each area under the standard normal curve using your standard normal table. The shaded graph shows the area. Start with the example below. The same problems are also in this Google Sheet.

Example: Using the Table

Find the left tail area for $z = -1.25$.



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611

Split $z = -1.25$ into the **row** -1.2 (the ones digit and first decimal place) and the **column** .05 (the second decimal place). Where they meet:

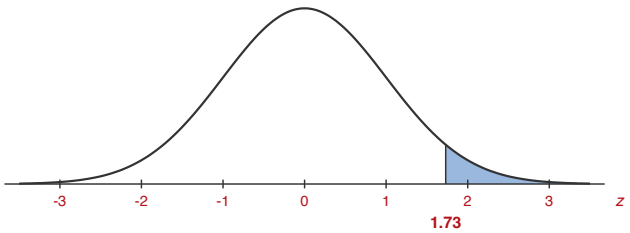
$$\text{left tail area} = 0.1056 \approx 11\%$$

Our table has only **left tail areas**. For the other areas, look up left tail areas first:

- **Right tail area** $= 1.0 -$ left tail area
- **Area between** z_1 and z_2 $=$ (left tail area for z_2) $-$ (left tail area for z_1)

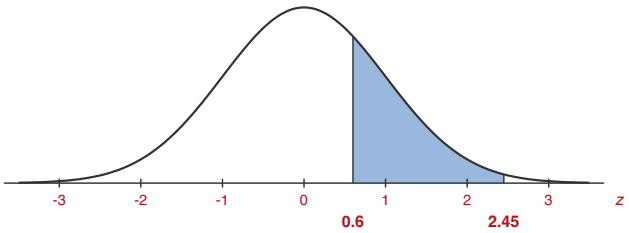
Problems

1. Find the right tail area for $z = 1.73$



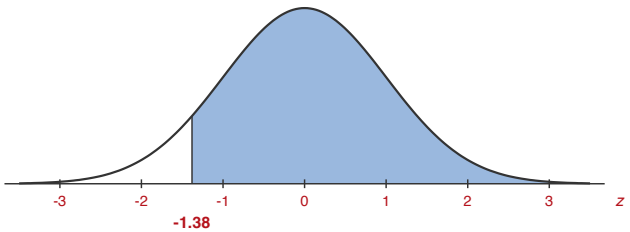
Area $=$ _____

2. Find the area between $z = 0.6$ and $z = 2.45$



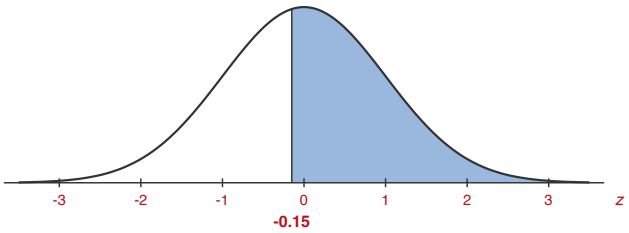
Area $=$ _____

3. Find the area to the right of $z = -1.38$



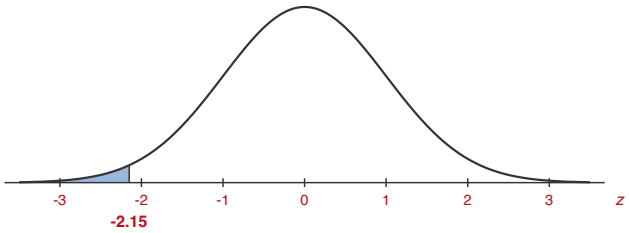
Area $=$ _____

4. Find the area where $z > -0.15$



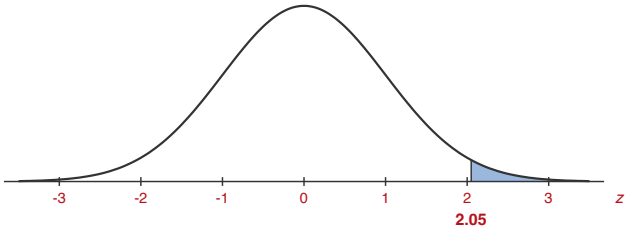
Area $=$ _____

5. Find the left tail area for $z = -2.15$



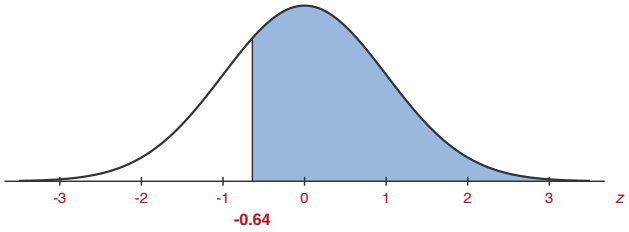
Area = _____

6. Find the area for z greater than 2.05



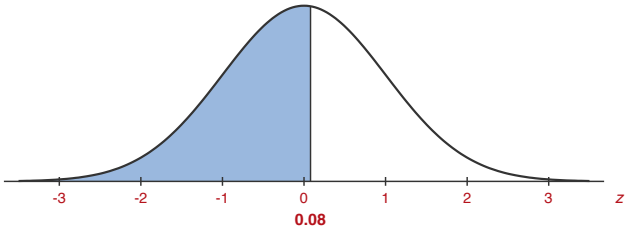
Area = _____

7. Find the right tail area for $z = -0.64$



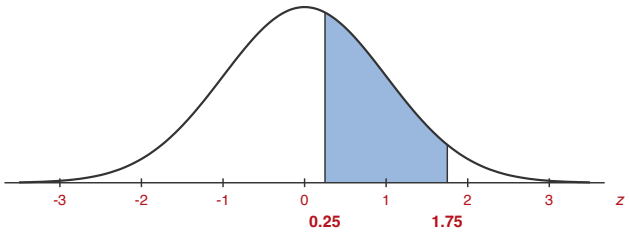
Area = _____

8. Find the area to the left of $z = 0.08$



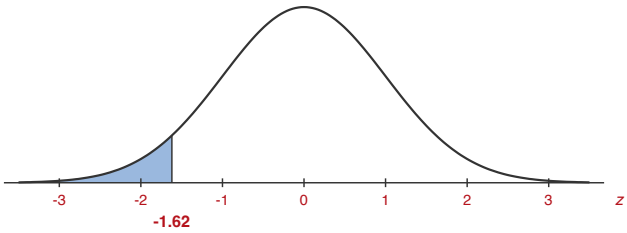
Area = _____

9. Find the area where $0.25 < z < 1.75$



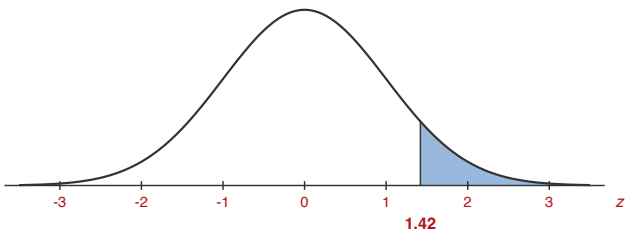
Area = _____

10. Find the area where $z < -1.62$



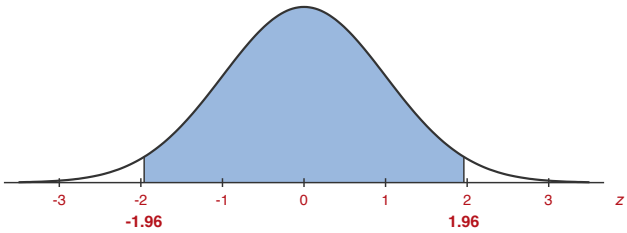
Area = _____

11. Find the area to the right of $z = 1.42$



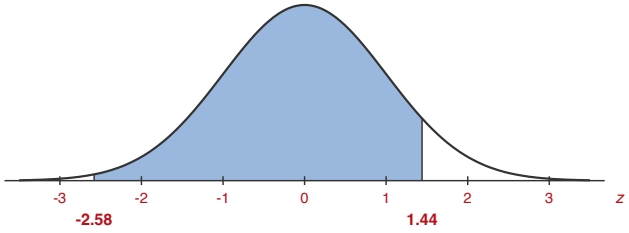
Area = _____

12. Find the area for z between -1.96 and 1.96



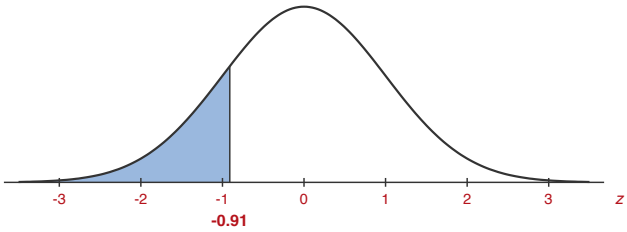
Area = _____

13. Find the area for z greater than -2.58 and less than 1.44



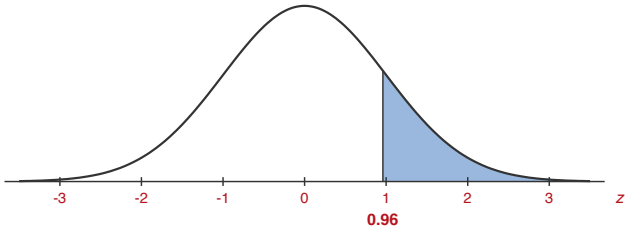
Area = _____

14. Find the area for z less than -0.91



Area = _____

15. Find the area where $z > 0.96$



Area = _____